UNIVERSITY OF JOHANNESBURG

SQL Case Statement

Lecturer's remarks
Docente Kommentar

Questions 1 Table: Products

Product_id	Product_name	Price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

Classify each product by price:

* Expensive if price > 1000
* Mid-range if price between 100 and 1000
* Budget if price < 100

```
SELECT product_name, price,
  Case
    WHEN price > 1000 THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000
      THEN 'Mid-range'
    ELSE 'Budget'
  END AS price_Category
FROM Products;
```

Product_name	Price	Price_Category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

Orders

order_id	Customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

Label each order:

High Value for orders ≥ 1000

Medium Value for 500-999.99

Low Value for orders < 500

```
SELECT Customer_name, amount,
  CASE
    WHEN amount >= 1000 THEN 'High_Value'
    WHEN amount BETWEEN 500 AND 999.99
      THEN 'Medium Value'
    ELSE 'Low Value'
  END AS order_value_Category
FROM orders;
```

Customer_name	amount	order_value_Category
Alice	150.00	Low value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

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Employees

Emp_id	Emp_name	department	Salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

Categorize employee position

If in 'IT' and Salary >80000 -> 'Senior IT'
If in 'HR' and Salary >55000 -> 'Experienced HR'
Otherwise -> 'Staff'

```
SELECT emp_name, department, Salary,
    CASE
      WHEN department = 'IT' AND Salary>80000
      THEN 'Sinior IT'
      WHEN department = 'HR' AND Salary>55000
       THEN 'Experienced HR'
    ELSE 'Staff'
    END AS position_level
FROM employees;
```

Emp_name	depart	Salary	Position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experience HR
Mark	IT	75000	Staff
Lucy	Finace	95000	Staff
Tom	HR	55000	Staff

Students

Student_id	Student_name	Score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

Assign a latter grade

≥ 90 'A'
80-89 'B'
70-79 'C'
60-69 'D'
< 60 'F'

```
SELECT Student_name, Score,
    CASE
      WHEN Score >=90 THEN 'A'
      WHEN Score >=80 THEN 'B'
       WHEN Score >=70 THEN 'C'
       WHEN Score >=60 THEN 'D'
    ELSE 'F'
    END AS grade
FROM Students;
```

Student_Name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

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Deliveries

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

Lable delivery performance

≤ 30 mins 'Fast'
31-60 mins 'On Time'
60 mins 'Late'

```
SELECT delivery_id, delivery_time_minutes,
    CASE
      WHEN delivery_time_minutes <=30 THEN 'Fast'
      WHEN delivery_time_minutes <=60 THEN 'on Time'
    ELSE 'late'
    END AS performance
FROM deliveries;
```

deliveg_id	delivery_time_min	Permance
1	45	onTime
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

Tickets

ticket_id	issue_type	Priority
1	Long-issue	1
2	Serverdown	3
3	Slowsystem	2
4	Email error	2
5	Password reset	1

Convert priority to lables

3 - High
2 - Medium
1 - Low

```
SELECT issue_type, priority,
    CASE priority
      WHEN 3 THEN 'High'
      WHEN 2 THEN 'Medium'
      WHEN 1 THEN 'Low'
    END AS priority_label
FROM tickets;
```

issue_type	Priority	Priority_label
Long issue	1	Low
Sever down	3	High
Slow System	2	Midium
Email Error	2	Medium
Password Rest	1	Low

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Attendance

Student_id	days_Present	total_day
1	45	50
2	30	50
3	48	50
4	28	50
5	50	50

Calculate attendence % and Classify

≥90% -> Excellent

75-89% -> Good

< 75% -> 'Needs improvement'

```
SELECT Student_id,
        (days_present *100.0/total_days) AS
        attendance_percentage,
CASE
 WHEN (days_present *100.0/total_days)>=90
  THEN 'excellent'
 WHEN (days_present*100.0/total_days)>75
  THEN 'Good'
  ELSE Needs Improvement
END AS Attendance_Status
FROM attendance;
```

Student_id	Attendance_Percentage	attendance_Status
1	90.0	Excellent
2	60.0	Needs improvement
3	96.0	Excellent
4	30.0	Needs improvement
5	100.0	Excellent

Products_inventory

Product_Id	Stock_qty
1	5
2	0
3	25
4	10
5	3

Lable Stock Status

0 -> out of Stock

1-5 -> Low Stock

5 -> in Stock

```
SELECT product_id, Stock_qty
  CASE
   WHEN Stock_qty =0 THEN 'out of Stock'
   WHEN Stock_qty <=5 THEN 'Low Stock'
   ELSE 'in_Stock'
  END AS Stock_Status
FROM products_inventory;
```

Product_id	Stock_qty	Stock_Status
1	5	Low Stock
2	0	out of Stock
3	25	IN Stock
4	10	inStock
5	3	Low Stock

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Classes

Class_Id	Subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

Class.79 by size

≥25 - Large
10-24 - Medium
< 10 - Small

```
SELECT Subject, enrolled_Students
    CASE
      WHEN enrolled_student >=25 THEN Large
      WHEN enrolled_Students >=10 THEN 'medium
    ELSE 'Small'
END AS Class_Size_Category
FROM Classes
```

Subject	Enrolled_Student	Class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

Payments

Payment_Id	Amount	Payment_Method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	PayPal
5	300.00	Cash

Apply discount Flag

If payment_Method = Cash and amount ≥200 →
Eligible for Discount
Otherwise → not Eligible

```
SELECT payment_id, payment_Method amount,
    CASE
      WHEN payment_method = 'Cash' AND amount
        >=200 THEN Eligible for Discount
      ELSE 'not Eligible'
END AS discount_elisibility
FROM payment
```

Payment_id	Payment_met	Amount	Discount Eligible
1	Card	50.00	not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	not Eligible
4	PayPal	75.00	not Eligible
5	Cash	300.00	Eligible for discount

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